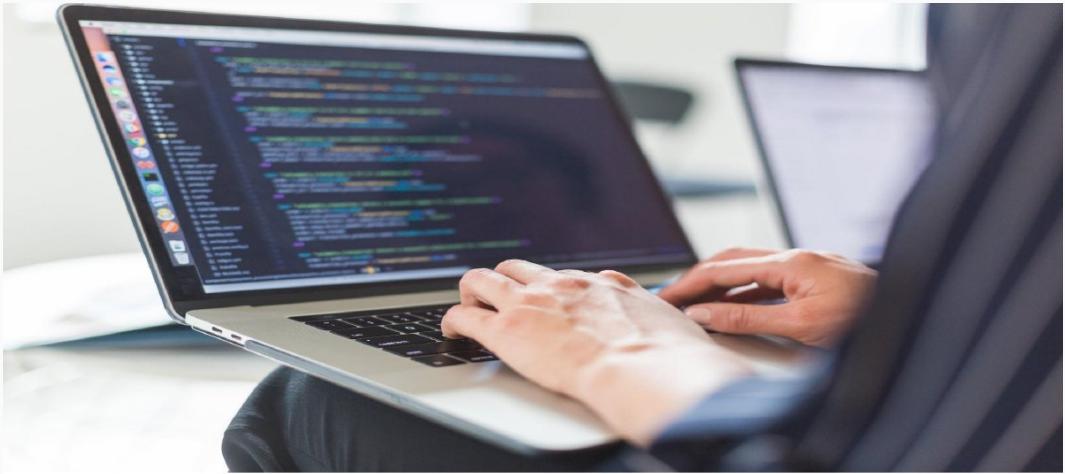


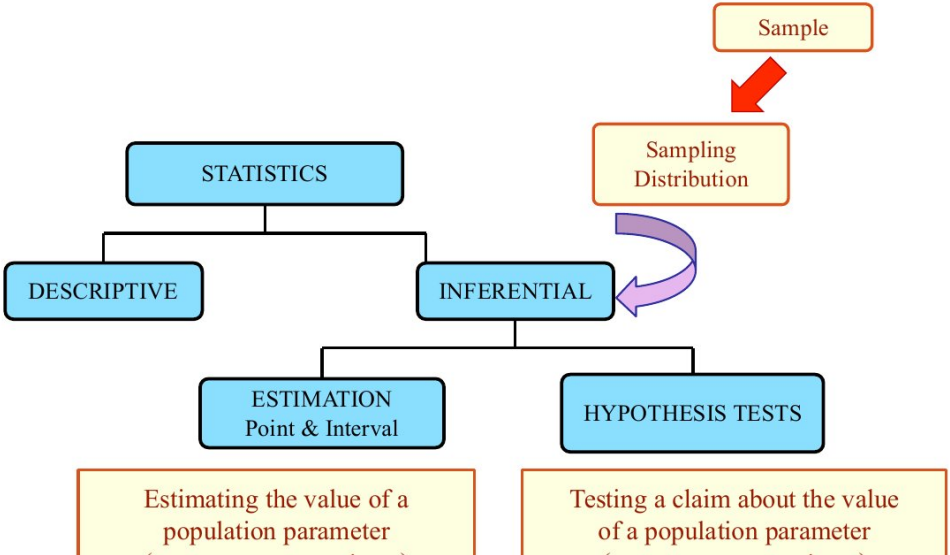
ETX1100/ETX5900 Business Statistics
Week 5
Hypothesis Testing II

Reminders

Assignment due on 19 Jan 2026 at 11:55pm.
Please start early!

Consultation (dedicated student hours) – see Moodle for time/location.

Hypothesis Test for Evidence-based Decisions



Lesson outcomes:

1. Identify a business problem
2. Formulate a hypothesis using variables relevant to the problem
3. Establish the degree of evidence for the hypotheses
4. Interpret results in the context of the business problem
5. Identify potential errors in hypothesis testing
6. Explain the implications of these errors for business decision-making

Textbook references:
Berenson et al *Basic Business Statistics 5th edition,*
Chapter 9, Sections 9.1-9.7

Hypothesis Testing – Classic Example

An hypothesis test is conducted to decide between two hypotheses: the null H_0 and the alternative, H_1.



H_0: Defendant is not guilty ('status quo'; default)

H_1: Defendant is guilty (experimental/research hypothesis)

Reject H_0: GUILTY!

There ***IS*** enough evidence against H_0 - the defendant is guilty.

Do not reject H_0: NOT GUILTY ~~**(INNOCENT!)**~~

There ***IS NOT*** enough evidence against H_0 - the defendant is not guilty.

5 Step Process in Hypothesis Testing

Step 1: Set up the null and the alternative hypotheses based on a benchmark value of a population parameter (μ_0 or π_0)

Note:
Test for a mean *(μ_0) when the data are* ***numerical measurements.***
Test for a proportion *(π_0) when the data are* ***yes/no or category outcomes.***

H_0: A default statement/status quo
(contains the equals sign: =)

H_1: A belief/claim that we aim to test
(what we are searching evidence for, does not contain the equals sign: ≠, >, <)

Step 2: Test Statistic

Compute Z_{calc} or t_{calc} *(depends on whether the population standard deviation (σ) is known or unknown).*

Calculating the test statistic for the mean:

$$Z = \frac{X - \mu}{\sigma/\sqrt{n}}$$ *the formula used when the population standard deviation σ is known*

$$t = \frac{X - \mu}{s/\sqrt{n}}$$ *the formula used when the sample standard deviation s is used (in most cases)*

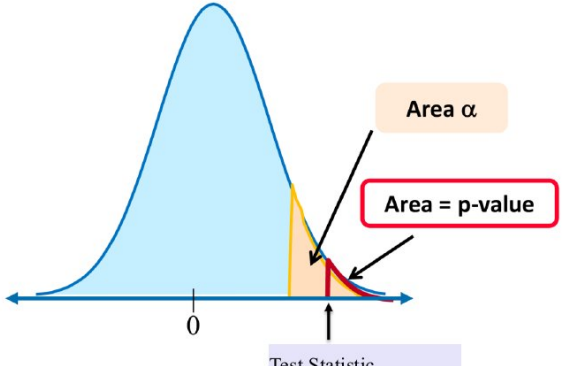
Calculating the test statistic for the proportion:

$$Z = \frac{p - \pi_0}{\sqrt{(\pi_0(1-\pi_0)/n)}}$$

Step 3: p value

Determine the p-value by using the calculated test statistic. Find the appropriate tail area (according to H_1)

The p-value determines the **area (p-value)** in the tail beyond the ***calculated*** test-statistic and compares it to **α**, the area in the tail associated with the given significance level.



Mechanics of Hypothesis Testing: p-value

Judging whether or not the test statistic is outstandingly "**far from zero**" in the direction of the alternative.

It is the **probability** of getting the test statistic **assuming that H_0 is true**.

p-value = P(Test statistic $|H_0$ is true)

If the p-value is small, then the test statistic is far from zero.
(In the direction of the alternative hypothesis.)



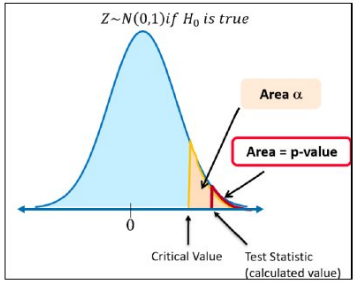
A smaller p-value means that there is stronger evidence in favor of the alternative hypothesis.

Interpreting the p-value

P-value should be interpreted as the level of "**evidence**" in favour of the null hypothesis, when compared against the ALTERNATIVE hypothesis.

A *small* p-value:
Indicates little evidence in support of the H_0;
we **have reason to reject H_0** in favour of H_1

A *large* p-value:
Indicates considerable evidence in support of the H_0;
we **do not have reason to reject H_0** in favour of H_1.



Advantage:
An advantage of p-values is that we can perform hypothesis tests at all significance levels without finding a new critical value.

Steps 4 and 5

Step 4: Decision Rule and Decision: We either reject H_0 or do not reject H_0

Reject H_0 if p-value < α
Don't reject H_0 if p-value > α
(Note: α is usually set at 1%, 5% or 10%. The convention is 5%)

Step 5: Conclusion

State the conclusion within the context of the problem

P-value smaller than α

There is sufficient evidence at 5% level of significance that the population mean IS ……(interpret H_1 here).

P-value larger than α

There is insufficient evidence at 5% level of significance that the population mean IS ……..(interpret H1 here).

Reject H_0 when p-value < α

If p-value < α, reject H_0

→ We still can tolerate the potential error and conclude that the p-value is small so we reject the H_0.

→ ***The smaller the p***-value, the **more evidence** there is in favour of H_1

∴ There is evidence to support the alternative hypothesis.

If p-value > α, fail to reject H_0

→ We no longer can tolerate the potential error as it is bigger than the tolerance level allowed, α. We conclude that the p-value is large so we fail to reject the H_0.

→ ***The larger the p***-value, the **less evidence** there is in favour of H_1

∴ There is not enough evidence to support the alternative hypothesis.

P-Value Approach - 5 Step Process

1. Hypotheses:

$H_0: \mu = \mu_0$
$H_1: \mu \neq \mu_0$

$H_0: \mu = \mu_0$
$H_1: \mu < \mu_0$

$H_0: \mu = \mu_0$
$H_1: \mu > \mu_0$

2. Calculate Test Statistic (Z_{calc} or t_{calc})
3. p-value
4. Decision Rule: Reject H_0 if p-value < α
5. Conclusion

1. Hypotheses:

$H_0: \pi = \pi_0$
$H_1: \pi \neq \pi_0$

$H_0: \pi = \pi_0$
$H_1: \pi < \pi_0$

$H_0: \pi = \pi_0$
$H_1: \pi > \pi_0$

2. Calculate Test Statistic (Z_{calc})
3. p-value
4. Decision Rule: Reject H_0 if p-value < α
5. Conclusion

Task 1: Marketing Campaign

The marketing team at an e-commerce company believes their email campaigns have an average click-through rate (CTR) of 12%. An analyst suspects that the CTR might be different from this benchmark. To investigate, the analyst randomly selects 45 recent email campaigns and finds an average CTR of 10.8% and standard deviation of 3.2%

Use a 5% significance level to test whether the true average CTR differs from 12%.

Task 1: Marketing Campaign

Let X = click through rate per campaign (%)

Since n=45 (>30), we can invoke CLT and assume that $\bar{X} \sim N(\mu, \sigma^2/n)$

1. Hypothesis

$H_0: \mu = 12$

$H_1: \mu \neq 12$

2. Test statistic

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$$

$$t_{calc} = \frac{10.8 - 12}{3.2/\sqrt{45}} = \frac{-1.2}{0.477028} = \mathbf{-2.51558}$$

Task 1: Marketing Campaign

Step 3: P-value

=P(t< - *2.51558*)*2

*= 0.007804*2 = 0.015609*

Since this is a two-tailed test:

*Excel function: = T.DIST(-2.51558,44,true)*2*

OR = T.DIST.2T(2.51558,44,true)

Task 1: Marketing Campaign

4. Decision rule:

Reject H_0 if p-value < α
Since 0.015609 < 0.05, we can reject H_0

5. Conclusion
There is sufficient evidence at the 5% level of significance that the true average CTR differs from 12%

Task 2: Online Transaction Success Rate

A bank states that **at least 95%** of online credit card transactions are processed without error. A financial analyst claims that this success rate may have dropped and wants to test if the current system is underperforming.

To investigate, the analyst reviews a **random sample of 250 online transactions**, and finds that **229 were processed without error**.

Is there evidence to support the financial analysts claim at the **10%** level of significance?

Task 2: Online Transaction Success Rate

Let X = number of online transactions

X=229 n=250 p= 229/250 = **0.916**

$n\pi$ = (250)(0.95) = 237.5 $n(1-\pi)$=(250)(1-0.95) = 12.5

The assumption of normality is reasonable since they are both ≥5.

Task 2: Online Transaction Success Rate

Step 1: Hypotheses

H_0: $\pi = 0.95$

H_1: $\pi < 0.95$

Step 2: Test Statistic

$$Z = \frac{p - \pi_0}{\sqrt{(\pi_0(1 - \pi_0) / n)}}$$

$$Z_{calc} = \frac{0.916 - 0.95}{\sqrt{(0.95(1 - 0.95) / 250)}} = -2.46662$$

Step 3: p-Value:

$=P(Z< -2.46662) = 0.00682$

Excel function: =NORM.S.DIST(-2.46662,true)

Task 2: Online Transaction Success Rate

Step 4: Decision Rule and Decision

Reject H_0 if p-value < α

Since 0.00682< 0.10, we ***can*** reject H_0

Step 5: Conclusion

There is sufficient evidence at the 10% level of significance that the true success rate of online credit card transactions **is lower than 95%.**

There is evidence to support the financial analyst's claim.

Task 2: Online Transaction Success Rate

b. Interpret the p-value in the context of the question.

If the true proportion were 0.95, then 0.682% of the time we would expect to get a sample proportion at least as small as the one we actually got (0.916).

Errors in Hypothesis Testing

- Type I Error is directly related to the significance level:

$$\Pr(\textit{Type I Error}) = \alpha$$

- So, we can control the chance of this error by setting small α
- Small $\alpha \rightarrow$ we need more convincing evidence against H_0 to reject it
- The smaller we make α, the bigger β becomes.

Type I and II errors

Since the conclusions that we draw are based on sample data - using a decision rule, there is the chance of making two possible errors:

Type I Error:

Reject H_0 when it's actually true.

Conclude guilty, when defendant is innocent.

- The probability of a Type I error is denoted by **α**.
- This value is also referred to as the ***"Level of Significance"***.

Type II Error:

Do not reject H_0 when it's actually false.

Conclude not guilty, when defendant is guilty.

- The probability of making a Type II error is denoted by **β**.
- **(1- β)** is referred to as the ***"Power of the Test".***

NOTE:

If we retain the null, we do not say it is true. We do not say we "accept" it; we simply do not reject it.

Type I and II errors

The smaller we make α, the bigger β becomes.
This inverse relationship must be considered when choosing the significance level α.
(1-β) is referred to as the ***"Power of the Test".***



The true 'state of the world'

Statistical Decision	H_0 TRUE	H_0 FALSE
Do not reject H_0	**Correct decision**	**Type II error** **Pr(Type II error) = β**
Reject H_0	**Type I error** **Pr(Type 1 error) = α**	**Correct decision**

Task 3: Interpretation of Type I and II errors

A recent newspaper article reported that, for 21 year old students, the mean amount spent per month on discretionary items (entertainment, dining out, etc) is **at most $275**. A researcher claims this is not the case. He surveyed a random sample of forty 21 year old students and obtained a mean of $283, with a standard deviation of $17.85.

Formulate the hypotheses and interpret Type I and II errors in the context of the question.

Task 3: Interpretation of Type 1 and II errors

Hypotheses:

H_0: $\mu = 275$

H_1: $\mu > 275$

Type I error:

A Type 1 error is rejecting H_0, when H_0 is true.

We find sufficient evidence the mean amount spent per month on discretionary items is higher than $275, when, in fact, it is not.

Type II error:

A Type II error is retaining H_0, when H_0 is false.

We find insufficient evidence the mean amount spent per month on discretionary items is higher than $275 when, in fact, it is.

Task 4: Interpretation of Type I and II errors

The Glendale Steel Company manufactures steel bars. If the production process is working properly, it turns out steel bars with mean length of at least 855mm with a standard deviation of 65mm (as determined from engineering specifications on the production equipment involved).
Longer steel bars can be used or altered, but shorter bars must be scrapped. You select a sample of 25 bars and the mean length is 832mm. Assume that the length of steel bars is normally distributed. The production manager claims that the production equipment needs to be adjusted.

Formulate the hypotheses and interpret Type I and II errors in the context of the question.

Task 4: Interpretation of Type 1 and II errors

Hypotheses:

H_0: $\mu = 855$

H_1: $\mu < 855$

Type I error:

A Type 1 error is rejecting H_0, when H_0 is true.
We find sufficient evidence the mean length of steel bars is less than 855mm when in fact it is not.

Type II error:

A Type II error is retaining the null, when the null is false.
We find insufficient evidence the mean length of steel bars is less than 855mm when in fact it is.